Saintgits College of Engineering, Kottayam (Autonomous)

Department of Electronics Engineering

MICRO PROJECT REPORT

Course: *Microcontrollers and Computer Architecture*

Course Code: 24EVT2225

Smart Air Quality Monitor

An IoT-Enabled Real-Time Environmental Monitoring System

Using ESP32, MQ-135, DHT11, 16×2 LCD (I2C) & Buzzer

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1 Introduction

Air quality has emerged as one of the most critical environmental concerns of the modern era. Indoor and outdoor air pollution from industrial gases, vehicular emissions, and household sources leads to a range of health issues including respiratory diseases, cardiovascular complications, and long-term neurological damage. Monitoring ambient air quality in real time is therefore essential for early warning and preventive action.

The **Smart Air Quality Monitor** is an embedded IoT system designed to continuously measure harmful gas concentrations alongside temperature and humidity. The system uses an **ESP32 microcontroller** as its core processing unit, paired with an **MQ-135 gas sensor** for detecting pollutants such as CO_2, smoke, and ammonia, and a **DHT11 sensor** for ambient temperature and humidity. Readings are displayed on a **16×2 LCD screen** (via I2C interface), and a **buzzer** provides an audible alarm when pollution exceeds dangerous thresholds.

This project was developed as part of the Microcontrollers and Computer Architecture (24EVT2225) course at Saintgits College of Engineering.

1.1 Objectives

- To design an embedded air quality monitoring system using the ESP32 microcontroller.
- To measure gas concentration (in PPM), temperature, and humidity in real time.
- To classify air quality into three levels: *Good*, *Moderate*, and *Dangerous*.
- To provide an audible buzzer alarm when pollution crosses the danger threshold (>300 PPM).
- To display live sensor data on a 16×2 I2C LCD screen.
- To implement a warm-up phase for the MQ-135 sensor ensuring accurate readings.

2 System Architecture and Functional Block Diagram

2.1 Functional Block Diagram

2.2 Explanation

The **MQ-135 sensor** generates an analog voltage proportional to gas concentration. The ESP32's ADC reads this on GPIO 34. The **DHT11 sensor** outputs temperature and humidity data digitally on GPIO 4. The ESP32 processes both sensor inputs, classifies air quality, and drives two output systems:

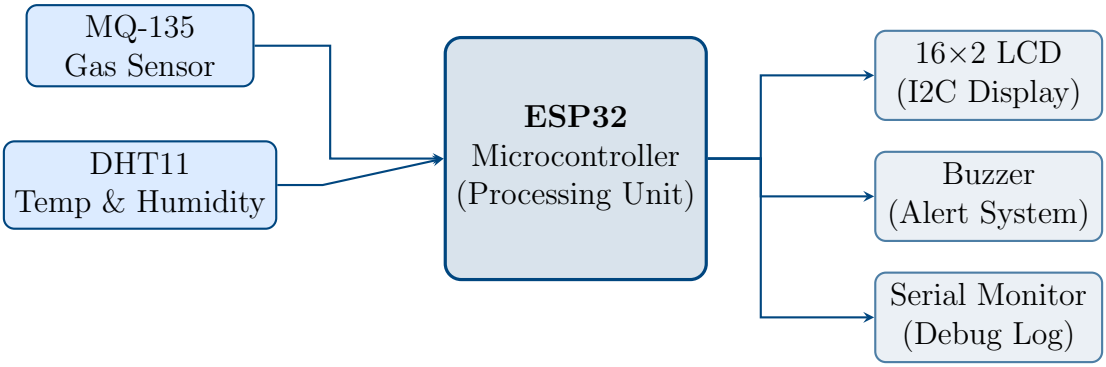

Figure 1: Functional Block Diagram of the Smart Air Quality Monitor

- A **16×2 I2C LCD** (address 0x27) that alternates between showing PPM/temperature and humidity/raw ADC readings every 3 seconds.
- A **buzzer** on GPIO 18 that intermittently beeps (200 ms ON / 800 ms OFF) when air quality reaches the DANGER level.

3 Circuit Diagram and Pin Interfacing

3.1 Circuit Diagram

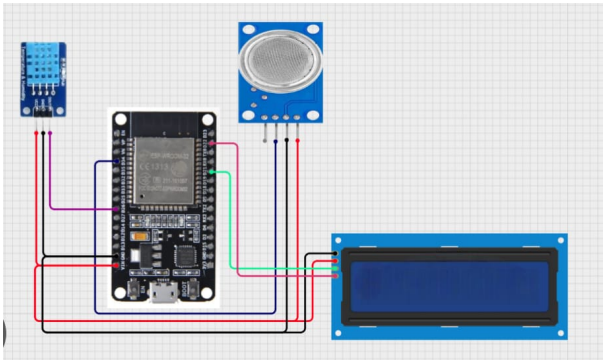

Figure 2: Circuit Diagram

3.2 Working Principle and Methodology

The system follows a non-blocking event-driven loop:

1. **Sensor Warm-Up:** On power-on, the MQ-135 is allowed a 2-minute warm-up period. The LCD counts down the remaining time.
2. **ADC Averaging:** The MQ-135 analog signal is sampled 8 times per cycle with 500 µs delays, and the average is used to reduce noise.

3. **PPM Mapping:** The averaged raw ADC value is baseline-subtracted and mapped linearly to a 0–1000 PPM range using the ESP32's `map()` function.
4. **DHT11 Reading:** Temperature and humidity are read every 2 seconds (DHT11 minimum interval).
5. **Air Quality Classification:** PPM is classified into three zones (see Table 1).
6. **LCD Display:** Two screens alternate every 3 seconds — Screen A shows PPM and temperature, Screen B shows humidity and raw ADC.
7. **Buzzer Alert:** In DANGER mode, the buzzer beeps in a 200 ms ON / 800 ms OFF pattern using non-blocking timing.
8. **Serial Logging:** Full sensor data is logged every 6 seconds for debugging.

4 Threshold Levels and Air Quality Classification

Table 1: Air Quality Threshold Levels

Category	PPM Range	Buzzer	LCD Label
Good	0 – 150	OFF	`GOOD`
Moderate	151 – 300	OFF	`MOD`
Dangerous	> 300	ON (beep)	`DANGER!`

5 Expected and Observed Outcomes

5.1 Working Model

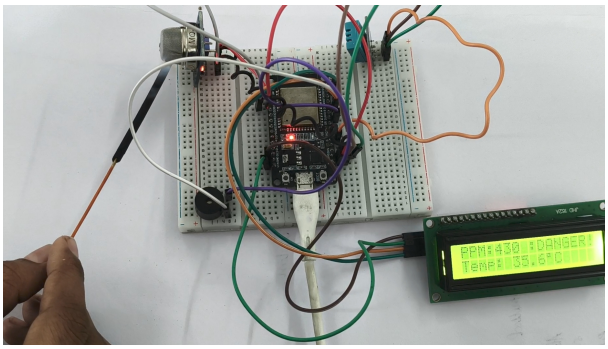

Figure 3

5.2 Expected Outcomes

- Real-time air quality (PPM), temperature, and humidity displayed on the LCD.
- Correct classification of air quality into GOOD, MODERATE, and DANGER categories.
- Audible buzzer alarm activates only when PPM exceeds 300.
- Non-blocking operation ensuring all sensors and outputs run simultaneously.
- Accurate PPM estimation after the 2-minute warm-up phase.

5.3 Observed Outcomes

- The MQ-135 successfully detects gas changes (e.g., incense smoke, matchstick fumes) and PPM rises accordingly.
- DHT11 provides stable temperature and humidity readings at 2-second intervals.
- LCD alternates between two information screens every 3 seconds without flicker.
- Buzzer produces distinct intermittent beeping only in the DANGER zone; remains silent otherwise.
- Serial monitor logs confirm correct sensor values and air quality classifications.

6 Conclusion

The Smart Air Quality Monitor successfully demonstrates the integration of sensor interfacing, real-time data processing, and output actuation on the ESP32 platform. The system classifies ambient air quality into three actionable levels and provides both visual (LCD) and audible (buzzer) feedback without the need for cloud connectivity, making it cost-effective and deployable in offline environments such as classrooms, homes, and offices.

The use of non-blocking timing (`millis()`-based) ensures all peripherals — sensor readings, display updates, and buzzer control — operate concurrently and efficiently. The 2-minute warm-up sequence and 8-sample ADC averaging significantly improve the reliability of MQ-135 readings.

This project provides a solid foundation for future extensions such as Wi-Fi-based cloud logging, mobile app notifications, and multi-gas sensor fusion.

7 References

1. K. Janaki, M. Eapen, M. Meera, M. A. Bhendale, S. Arun, and R. Krishnamoorthy, "Enhancing Environmental Safety: Design and Development of IoT based Air Quality Monitoring System," *2024 5th International Conference on Image Processing and Capsule Networks (ICIPCN)*, pp. 845–850, IEEE, 2024.

2. M. Jouda and M. Wadi, "IoT With LoRa Architecture For Indoor Air Quality Monitoring System," *8th International Artificial Intelligence and Data Processing Symposium (IDAP)*, IEEE, 2024.

3. Espressif Systems, *ESP32 Technical Reference Manual*, v5.1, Espressif Systems, 2023. Available: https://www.espressif.com/sites/default/files/documentation/esp32_technical_reference_manual_en.pdf

4. Adafruit Industries, *DHT11 Humidity & Temperature Sensor Datasheet*, Adafruit, 2022. Available: https://learn.adafruit.com/dht

5. Winsen Electronics, *MQ-135 Gas Sensor Datasheet*, Zhengzhou Winsen Electronics Technology Co., Ltd., 2014.